

Shanmukha Srinivas Challa

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PROFESSIONAL SUMMARY

Software Engineer and MS Data Science candidate (NJIT, GPA 3.8) with 4 years of hands-on experience building AI/ML systems, data pipelines, and full-stack applications — including RAG assistants, multi-agent pipelines, and production REST APIs. Reduced API latency by processing 1M+ records in ~3 seconds and achieved 90%+ citation coverage on a medical RAG evaluation set. Proficient in Python, FastAPI, React, PostgreSQL, TensorFlow, PyTorch, Docker, and AWS — open to AI/ML Engineering, Data Science, Data Engineering, and Full-Stack roles.

EXPERIENCE

Software Engineer Intern | Virtusa, Hyderabad, India

Jan 2024 – Jun 2024

- Designed and optimized C# / ASP.NET Core REST APIs to process 1M+ records in ~3 seconds, enabling near-real-time batch analytics.
- Refactored React UI components into reusable patterns and shared utilities, reducing front-end development effort by 30%.
- Implemented CI/CD pipelines and deployment gates in Azure DevOps and Jenkins, cutting deployment time by 40% and improving release reliability.

Software Engineer Intern | Sanjana Logistics LLP, Visakhapatnam, India

May 2023 – Jul 2023

- Optimized Oracle SQL queries and indexing strategies, improving query execution speed by 45% for reporting and operational workloads.
- Built KPI dashboards and data analysis reports using React, Spring Boot, and Tableau, enabling trend analysis and improving decision turnaround by 15% through enhanced drill-down visibility for stakeholders.
- Migrated infrastructure to AWS EC2 and S3, improving scalability while reducing infrastructure costs by 20%.

PROJECTS

Full-Stack FastAPI Calculator | FastAPI, PostgreSQL, JWT, React, Alembic, Pytest, Playwright

- Built a JWT-authenticated CRUD web app with user-scoped workflows and export features (CSV/JSON/PDF), cutting manual reporting time by 50%.
- Achieved ~96% test coverage across 194 automated unit, integration, and end-to-end tests, significantly improving release confidence.
- Designed Alembic-managed schema migrations and role-scoped data access, ensuring zero-downtime deployments and secure multi-user isolation.

Medical Triage + RAG — Evidence-Based Q&A with Citations | Python, RAG, Embeddings, Vector Search, FastAPI, LangChain, OpenAI API

- Built a retrieval-augmented medical Q&A assistant delivering citation-grounded answers, achieving 90%+ citation coverage on evaluation set.
- Improved citation accuracy by 15% across 50 evaluation questions by optimizing ingestion, retrieval, and re-ranking pipelines.
- Trained and evaluated a FAISS-based vector search pipeline with semantic chunking and BERT re-ranking using cross-validation, reducing hallucination rate and improving answer relevance by 20%.

Full-Stack Banking Web App | Flask, SQL, Python, HTML/CSS

- Developed onboarding and transaction workflows with role-based admin access and audit-friendly history, reducing manual effort by 50%.
- Added input validation and workflow automation that improved end-to-end transaction processing time by ~35% and ensured full traceability.
- Engineered session management and CSRF-protected forms with real-time balance updates, supporting concurrent multi-user access without data conflicts.

AssedGuard — Multi-Agent Manufacturing Audit AI | Python, FastAPI, React, Multi-Agent Systems, RAG, Cognee, ReportLab

- Built a 4-agent pipeline (Scout → Ranker → Fixer → Narrator) that detects, ranks, auto-corrects, and explains corrupted manufacturing audit data, producing a signed-ready PDF narrative.
- Engineered a batch processing data pipeline with deterministic detection of 6 issue classes (duplicates, outliers, unit conflicts, compliance contradictions) using Cognee as a shared memory bus; runs fully offline.
- Delivered a non-technical compliance officer interface — drag CSV, auto-run agents, download corrected dataset and audit narrative — with zero engineering setup required.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript/TypeScript, C/C++, C#, HTML/CSS

Frameworks: FastAPI, Flask, Spring Boot, React, Angular, Node.js/Express, Next.js, Django; Pytest, JUnit

ML / AI: Deep Learning, NLP, Transformers (BERT, GPT, CLIP), TensorFlow, PyTorch, Scikit-learn, LightGBM, RAG, Embeddings, pandas, NumPy, statistical analysis, predictive modeling

MLOps / DevOps: Docker, Kubernetes, Jenkins, Azure DevOps, GitHub Actions, Terraform, Ansible, Packer, Tekton

Data / Cloud: ETL, Warehousing, Kafka, Pub/Sub, Airflow, Spark; PostgreSQL, MySQL, Oracle, MS SQL, MongoDB; AWS, Azure, GCP

Tools / BI: Git, Postman, Splunk, JIRA; Power BI, Tableau

EDUCATION

M.S. Data Science — New Jersey Institute of Technology, Newark, NJ

GPA: 3.8 | Aug 2024 – May 2026

B.Tech. Computer Science & Engineering — Gayatri Vidya Parishad College (A), India

GPA: 3.3 | Jul 2020 – Jul 2024

ACHIEVEMENTS & CERTIFICATIONS

Awards: 3rd Place – Hack-The-Mountain Hackathon (Fraud Detection: AWS + LightGBM) | Finalist – ENCRYPT 2022: Blockchain identity verification tool, reducing verification time by 35% | IEEE CS Organizing Lead; Youth for Sustainability Volunteer; Rotaract Member

Certifications: AWS ML Foundations | AWS Cloud Foundations | AWS Data Analytics | Research Publication: Impact of IoT on Education Sector After COVID-19 (Business & Management Journal, India)